From audio to a physical robot face

Timestamped execution, a differentiable motion surrogate, and constrained visual supervision.

a Timestamped physical execution

REAL ROBOT / RECORDED OBSERVATIONS

Speech audio
TTS / playback
Original waveform

→

Audio2Face
51D Noetix / ARKit
30 Hz source trajectory

→

Actual commands
Logged 3-to-1 selection
51D blendshapes at 10 Hz

→

Physical robot
Fixed vendor BS-to-motor
Mechanical facial response

→

Camera observation
Real face video + recorded audio
Actual per-frame timestamps

ONE HOST CLOCK Audio playback / A2F source frames / selected commands / camera frames

b Differentiable Robot-FLAME Simulator

SPEECH-v2 ADAPTED / FROZEN DOWNSTREAM

Predict robot geometry from actual commands, not from motor inversion.

Command history
12 steps x 51 channels
Measured interval phase

→

Calibrated prior + dynamics
Frozen static mapper → robot adapter
+ bounded temporal TCN residual

→

Expression + jaw
50D expression
3D jaw axis-angle

→

Fixed-identity FLAME2020
Shared MICA identity / canonical pose
5023-vertex robot-aligned geometry

REAL-VIDEO WEAK TARGET

MICA identity + EMOCA expression + DECA jaw/pose
Reconstructed from recorded robot video; used for Simulator supervision.

Hybrid reconstruction is a weak prior, not measured 3D ground truth.

c Evaluation and bounded visual-prior research

SEPARATE OBJECTIVES, NOT A SINGLE SOLVED SYSTEM

2D AUDIOVISUAL SYNCHRONIZATION

Full-sequence offset search

Audio + real camera video

Sliding windows → distance curve → best offset

Spatial-crop and domain-transfer controls.

3D AUDIO-MOTION SYNCHRONIZATION

Robot-SyncNet3D / guard

Aligned audio + Simulator mouth motion

Neutral-relative VOCA585 / canonical pose

Temporal guard; fine sync remains limited.

2D TEACHER / EXPLORATORY

Constrained articulation priors

LatentSync video → lipreading features

Rendered Robot-FLAME → feature comparison

Bounded mouth/jaw residuals; physical gains unproven.

Data / inference

Weak supervision

Architecture overview, not a turnkey training release.

LatentSync stays in 2D; it is NOT the current SyncNet3D positive-motion source.